

REPORT

Title

Prolog Program to Implement Backward Chaining

Aim

To implement **Backward Chaining** using facts, rules, and queries in Prolog.

Problem Statement

Implement a Prolog program that uses backward chaining to determine whether a given conclusion can be proved from a set of facts and rules.

Requirements

- SWI-Prolog
- Computer system
- Prolog source file (`exp6.pl`)

Procedure

1. Open **SWI-Prolog**.
2. Create a new file named `exp6.pl`.
3. Enter the Backward Chaining Prolog program.
4. Save the file.
5. Load the program using `[exp6]..`
6. Execute the required queries.
7. Observe whether Prolog returns `true` or `false`.

Program

```
% Facts
parent(john, mary).
parent(john, david).
parent(mary, susan).

male(john).
male(david).
female(mary).
female(susan).

% Rules
father(X, Y) :-
    parent(X, Y),
    male(X).
```

```
mother(X, Y) :-  
    parent(X, Y),  
    female(X).  
  
grandparent(X, Z) :-  
    parent(X, Y),  
    parent(Y, Z).  
  
grandfather(X, Z) :-  
    grandparent(X, Z),  
    male(X).  
  
grandmother(X, Z) :-  
    grandparent(X, Z),  
    female(X).
```

Sample Output

Query 1:

```
?- [exp6].  
true.
```

```
?- father(john, mary).  
true.
```

Query 2:

```
?- mother(mary, susan).  
true.
```

Query 3:

```
?- grandparent(john, susan).  
true.
```

Query 4:

```
?- grandfather(john, susan).  
true.
```

Query 5:

```
?- father(mary, susan).  
false.
```

Conclusion

Thus, the **Backward Chaining program was successfully implemented in Prolog** using facts, rules, and logical queries. Prolog starts from the given goal and works backward to verify whether the required facts and rules can prove the goal.